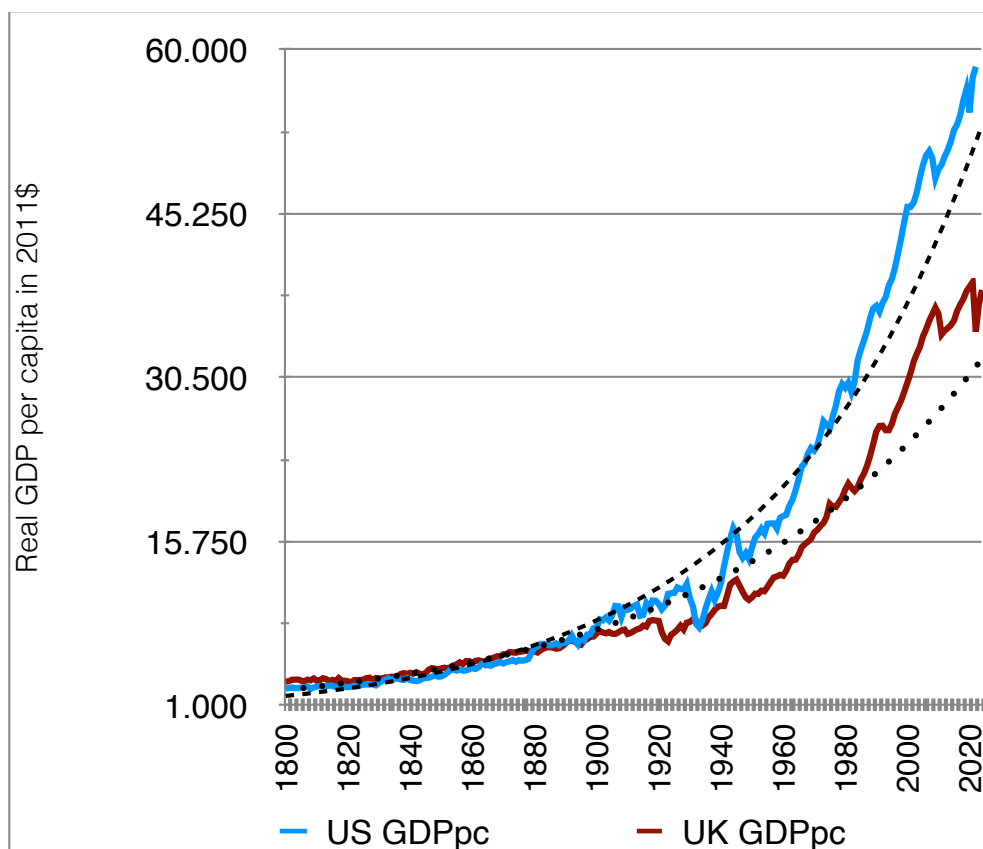


Economic Growth

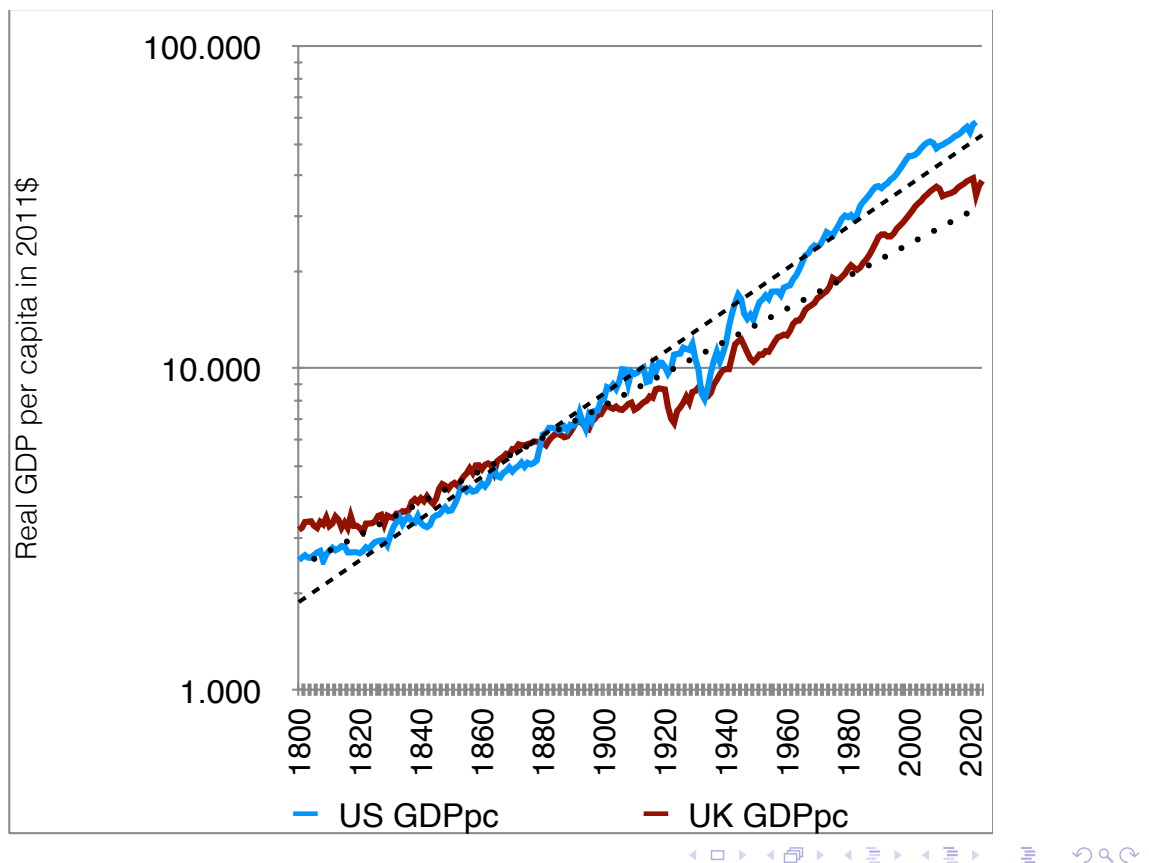
Pedro G. Duarte

INSPER

Mestrado Profissional em Economia



GDP per capita: US and UK (logarithmic scale)



Growth is a Recent Phenomenon!



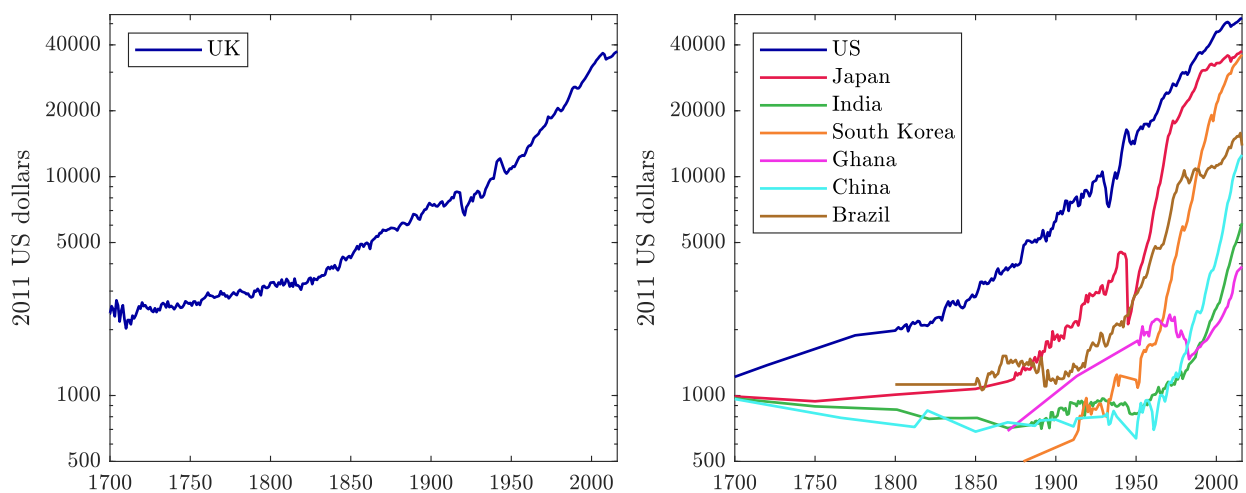
Real Wages of Laborers in England, 1250-2000. Source: Gregory Clark (2010); logarithmic scale

Economic Growth: the very long run

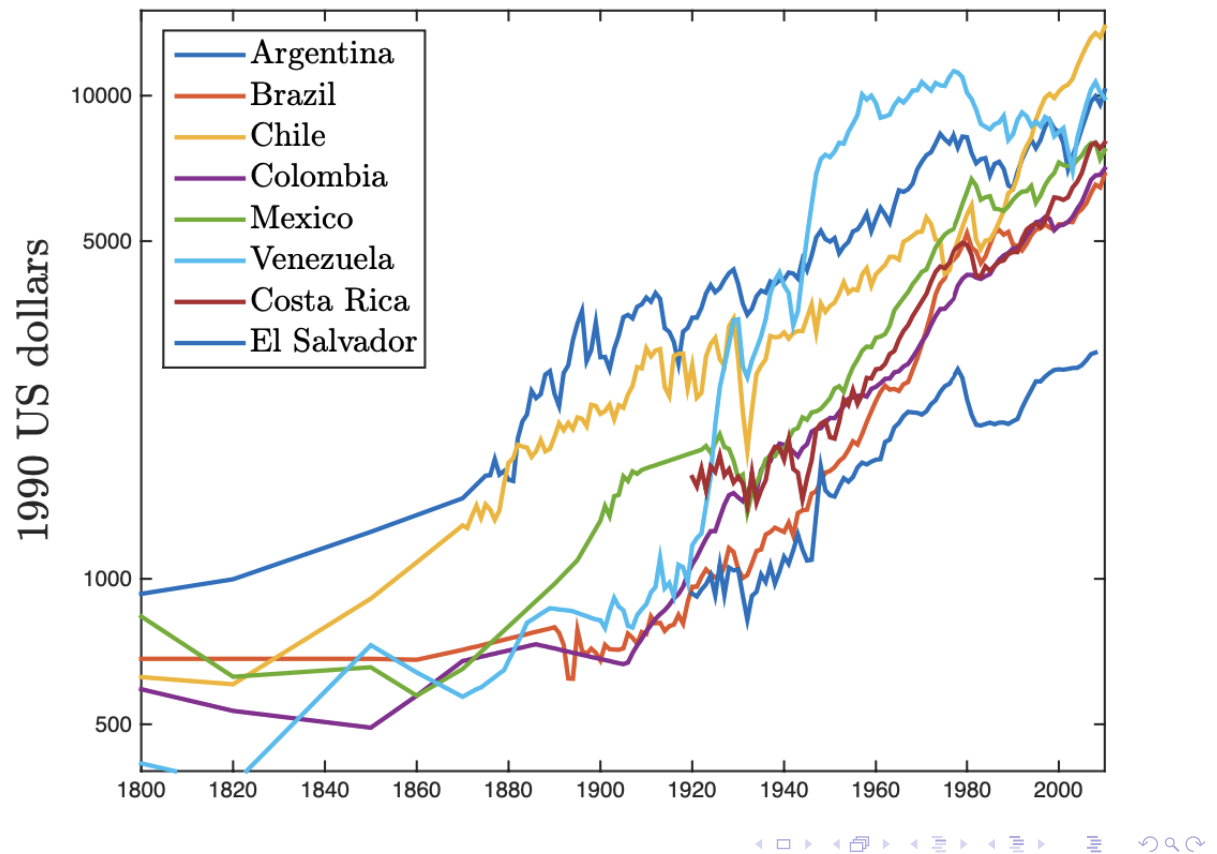
- Maddison (2001): build very long time series of GDP per capita for many countries
- No national accounts, but pieces of data:
 - ▶ people's average height
 - ▶ total number of livestock
 - ▶ crop yields
 - ▶ total output in specific industries (iron)
- Estimated subsistence level of consumption is around US\$ 400 per year at current prices
- UK has the best data and sustained growth since 19th century



Growth (GDP per capita): UK and Selected Countries



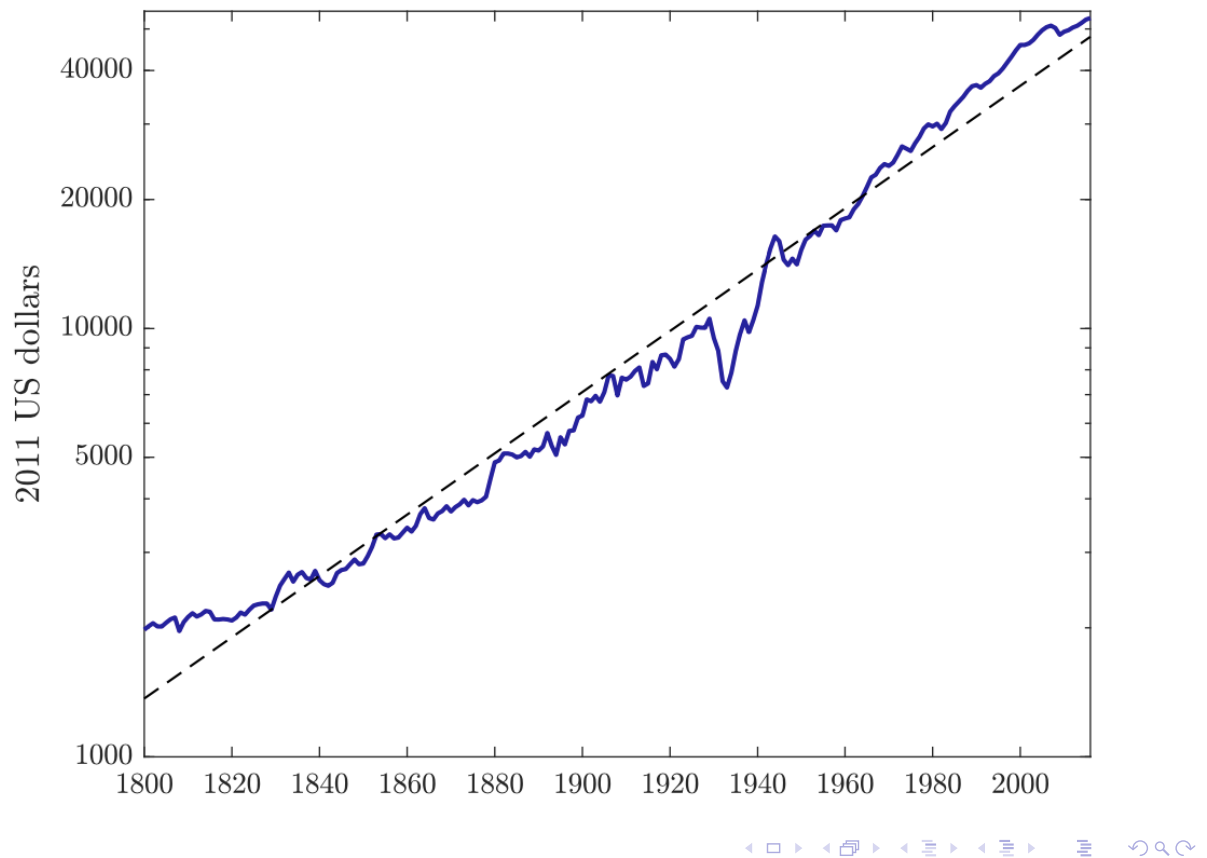
Latin America (logarithmic scale)



The Kaldor Facts

- Kaldor (1957): “remarkable historical constancies” in growing economies:
 - 1 Constant growth rate of GDP per capita

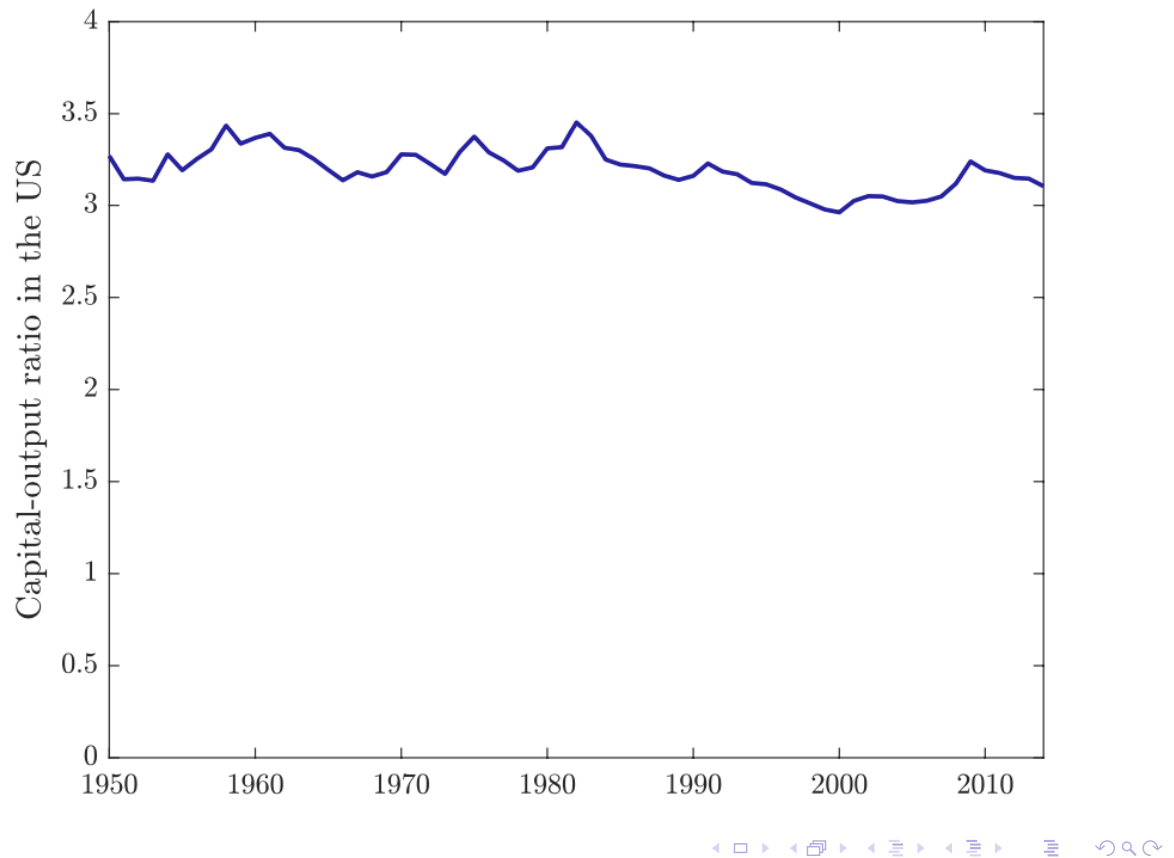
GDP per capita: US (logarithmic scale)



The Kaldor Facts

- Kaldor (1957): “remarkable historical constancies” in growing economies:
 - 1 Constant growth rate of GDP per capita
 - 2 Constant capital-output ratio

US Capital-Output Ratio



The Kaldor Facts

- Kaldor (1957): “remarkable historical constancies” in growing economies:
 - 1 Constant growth rate of GDP per capita
 - 2 Constant capital-output ratio
 - 3 Constant shares of labor and capital income in GDP

US Factor Shares

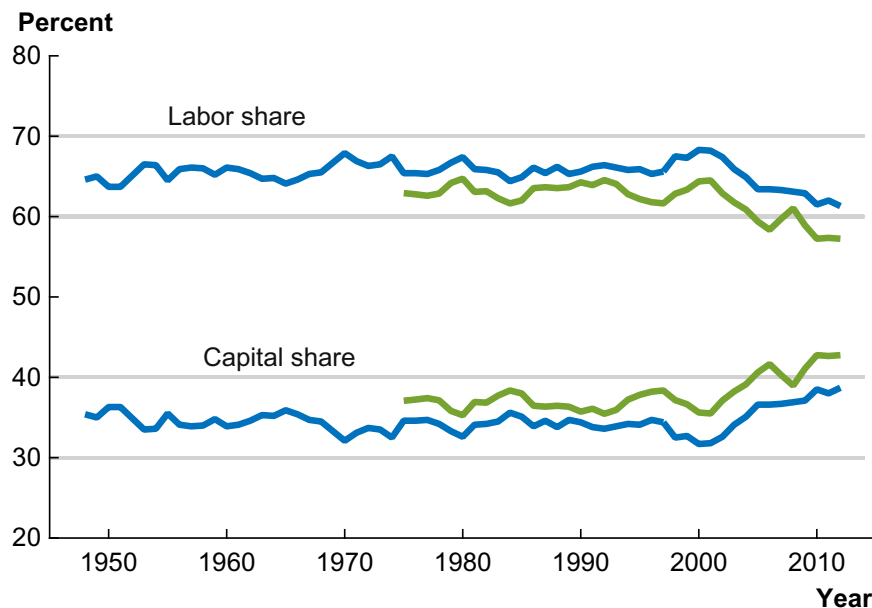


Fig. 6 Capital and labor shares of factor payments, United States. Source: *The series starting in 1975 are from Karabarbounis, L., Neiman, B. 2014. The global decline of the labor share. Q. J. Econ. 129 (1), 61–103. <http://ideas.repec.org/a/oup/qjecon/v129y2014i1p61-103.html> and measure the factor shares for the corporate sector, which the authors argue is helpful in eliminating issues related to self-employment. The series starting in 1948 is from the Bureau of Labor Statistics Multifactor Productivity Trends, August 21, 2014, for the private business sector. The factor shares add to 100%.*

Navigation icons: back, forward, search, etc.

The Kaldor Facts

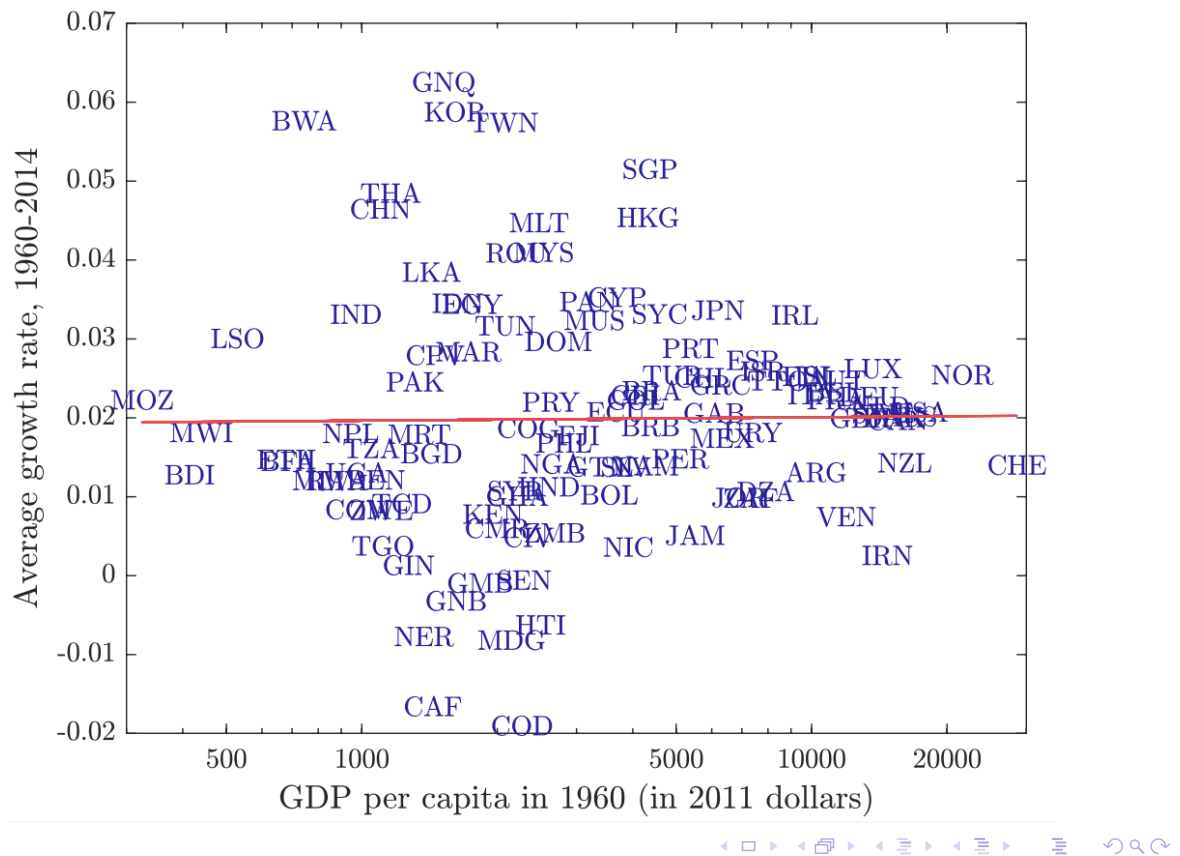
- Kaldor (1957): “remarkable historical constancies” in growing economies:

- ① Constant growth rate of GDP per capita
- ② Constant capital-output ratio
- ③ Constant shares of labor and capital income in GDP
- ④ Constant average rate of return on capital (implied by facts 2 and 3!):

$$\text{Return on Capital} = \frac{\text{Capital Income}}{\text{Capital Stock}} = \frac{\frac{\text{Capital Income}}{\text{GDP}}}{\frac{\text{Capital Stock}}{\text{GDP}}}$$

Navigation icons: back, forward, search, etc.

Average Growth by Country



Exercises

1 Suppose:

- ▶ The Capital-Output Ratio is 4
- ▶ The Capital Share in GDP is 0.2
- ▶ What is the (gross) rate of return on capital?

Exercises

① Suppose:

- ▶ The Capital-Output Ratio is 4
- ▶ The Capital Share in GDP is 0.2
- ▶ What is the (gross) rate of return on capital?

② Suppose

- ▶ The Capital-Output Ratio is rising over time
- ▶ The Capital Share in GDP is rising over time
- ▶ How does the (gross) return on capital change over time?



Big Picture Questions about Growth

- What sustains growth at the frontier?
(Will it continue in the future?)
- Why are some countries so far behind the frontier?
(What might help them close the gap?)
- Why did growth begin?
- Why was there no growth before Industrial Revolution?

We will focus on first two questions.



Is Capital Accumulation Key to Growth?

- Seems plausible!
- Conventional wisdom in 1950s: Yes!
- Solow (1956) tackled this question with a dynamic model (here: discrete time)



The Solow Model, 1: The Production Function

$$Y(t) = F[K(t), L(t)]$$

- $Y(t)$: Output at time t
- $K(t)$: Capital stock at time t
- $L(t)$: Labor supply at time t



The Solow Model, 1: The Production Function

$$Y(t) = F[K(t), L(t)]$$

Properties of the production function

① Constant Returns to Scale: $\lambda Y(t) = F[\lambda K(t), \lambda L(t)]$

② Positive Marginal Products:

$$F_K[K(t), L(t)] \equiv \frac{\partial F(K, L)}{\partial K} > 0$$

$$F_L[K(t), L(t)] > 0$$

③ Diminishing Marginal Products

$$F_{KK}[K(t), L(t)] \equiv \frac{\partial^2 F(K, L)}{\partial K^2} < 0$$

$$F_{LL}[K(t), L(t)] < 0$$

◀ ◻ ▶ ◀ ◻ ▶ ◀ ≡ ▶ ◀ ≡ ▶ ≡ ↺ ↻

The Solow Model, 1: The Production Function

$$Y(t) = F[K(t), L(t)]$$

Properties of the production function

④ Inada Conditions:

$$\lim_{K \rightarrow 0} F_K[K(t), L(t)] = \infty$$

$$\lim_{K \rightarrow \infty} F_K[K(t), L(t)] = 0$$

$$\lim_{L \rightarrow 0} F_L[K(t), L(t)] = \infty$$

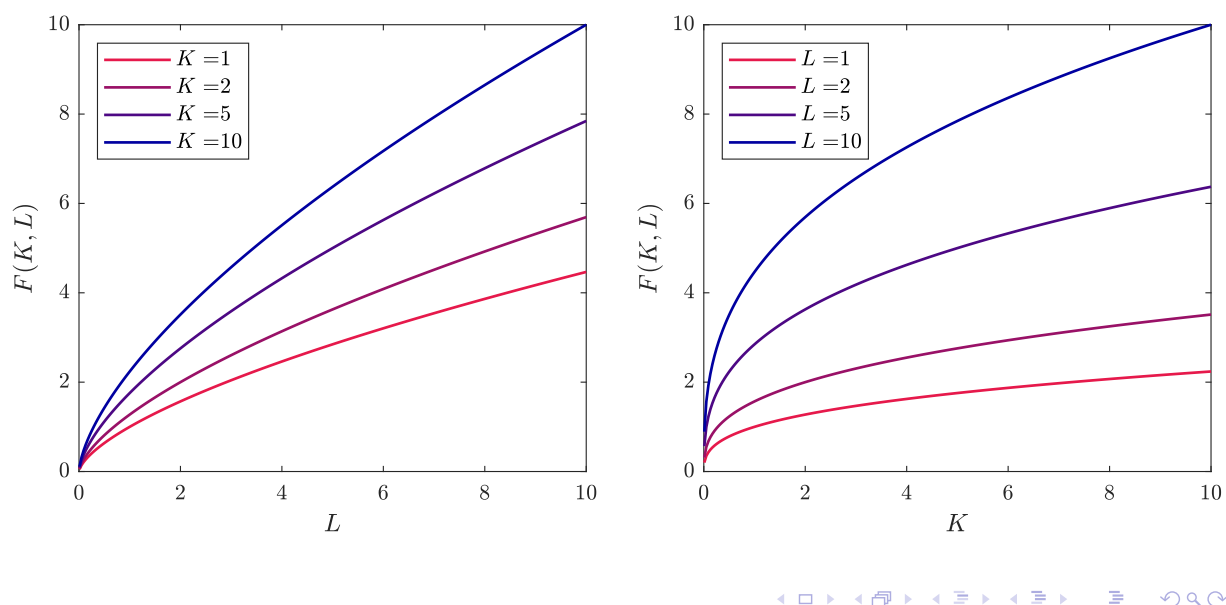
$$\lim_{L \rightarrow \infty} F_L[K(t), L(t)] = 0$$

◀ ◻ ▶ ◀ ◻ ▶ ◀ ≡ ▶ ◀ ≡ ▶ ≡ ↺ ↻

The Solow Model, 1: The Production Function

One function that satisfies these properties and is commonly used is the Cobb-Douglas:

$$Y(t) = K(t)^\alpha L(t)^{1-\alpha}$$



The Solow Model, 2: Population and Labor Supply

- We assume full employment, so $L(t)$ is both population and the labor force
- We assume an exogenous constant population growth:

$$L_{t+1} = (1 + n)L_t$$

The Solow Model, 3: Consumption and Investment

- Closed economy without government: $X = M = G = 0$

$$Y = C + I$$

- Exogenous and constant savings rate: $\frac{Y-C}{Y} \equiv \frac{S}{Y} = s$

- Therefore:

$$I(t) = S(t) = s.Y(t)$$



The Solow Model, 4: Depreciation and Capital Accumulation

- Constant depreciation rate, $\delta > 0$

$$K_{t+1} = (1 - \delta)K_t + I_t$$



The Mechanics of the Solow Model

- Define:

$$k_t \equiv \frac{K_t}{L_t}$$

- $$y_t \equiv \frac{F(K_t, L_t)}{L_t} = F\left(\frac{K_t}{L_t}, 1\right) \equiv f(k_t)$$



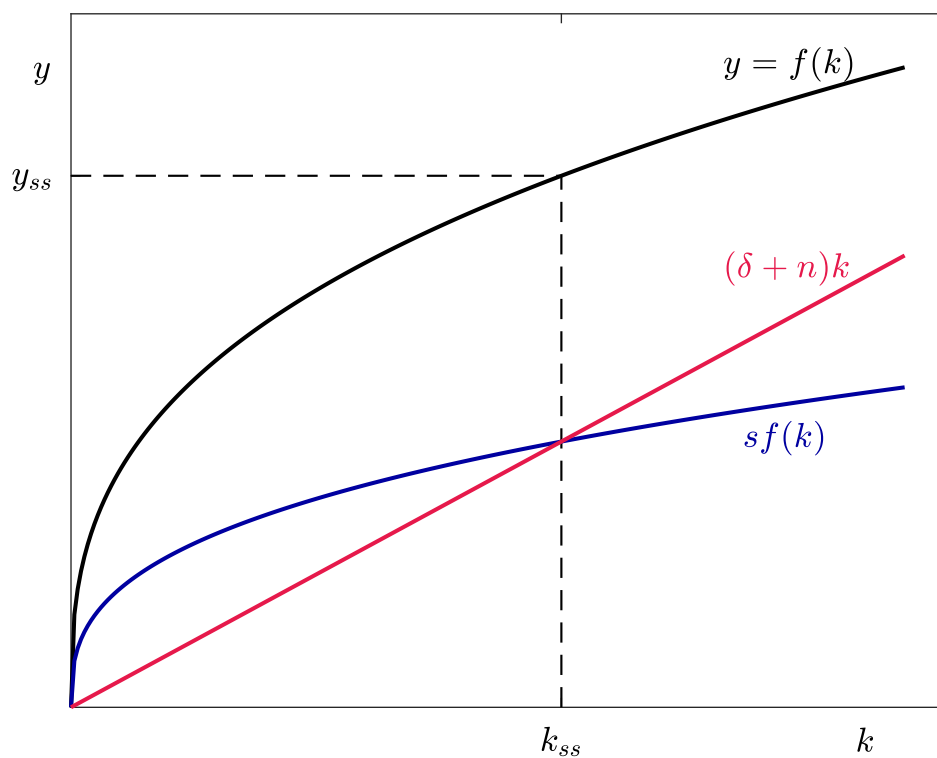
The Mechanics of the Solow Model

- ② Dynamics of capital per capita, Δk_{t+1}

$$\begin{aligned}
\Delta k_{t+1} &\equiv k_{t+1} - k_t = \frac{K_{t+1}}{L_{t+1}} - k_t \\
&= \frac{(1-\delta)K_t + I_t}{L_{t+1}} - k_t \\
&= \frac{(1-\delta)K_t + sY_t}{L_{t+1}} - k_t \\
&= \left[\frac{(1-\delta)K_t + sY_t}{L_t} \right] \frac{L_t}{L_{t+1}} - k_t \\
&= [(1-\delta)k_t + sy_t] \frac{1}{1+n} - k_t \\
&= \frac{sy_t - (\delta + n)k_t}{1+n} \\
&= \frac{sf(k_t) - (\delta + n)k_t}{1+n}
\end{aligned} \tag{1}$$



The Mechanics of the Solow Model



Navigation icons: back, forward, search, etc.

The Mechanics of the Solow Model

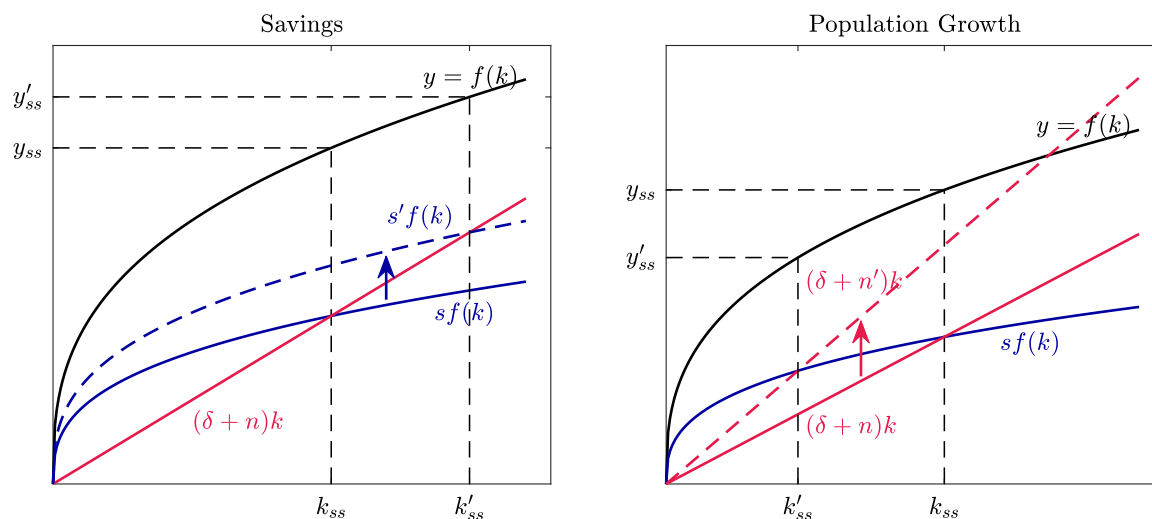
For the **Cobb-Douglas Production Function** we can compute k_{ss} and y_{ss} analytically:

$$\begin{aligned}
 y_t &= k_t^\alpha \\
 \Delta k_{t+1} &= \frac{sk_t^\alpha - (\delta + n)k_t}{1 + n} \\
 0 &= \frac{sk_{ss}^\alpha - (\delta + n)k_{ss}}{1 + n} \\
 k_{ss} &= \left(\frac{s}{\delta + n} \right)^{\frac{1}{1-\alpha}} \\
 y_{ss} &= \left(\frac{s}{\delta + n} \right)^{\frac{\alpha}{1-\alpha}}
 \end{aligned} \tag{2}$$

Navigation icons: back, forward, search, etc.

The Mechanics of the Solow Model

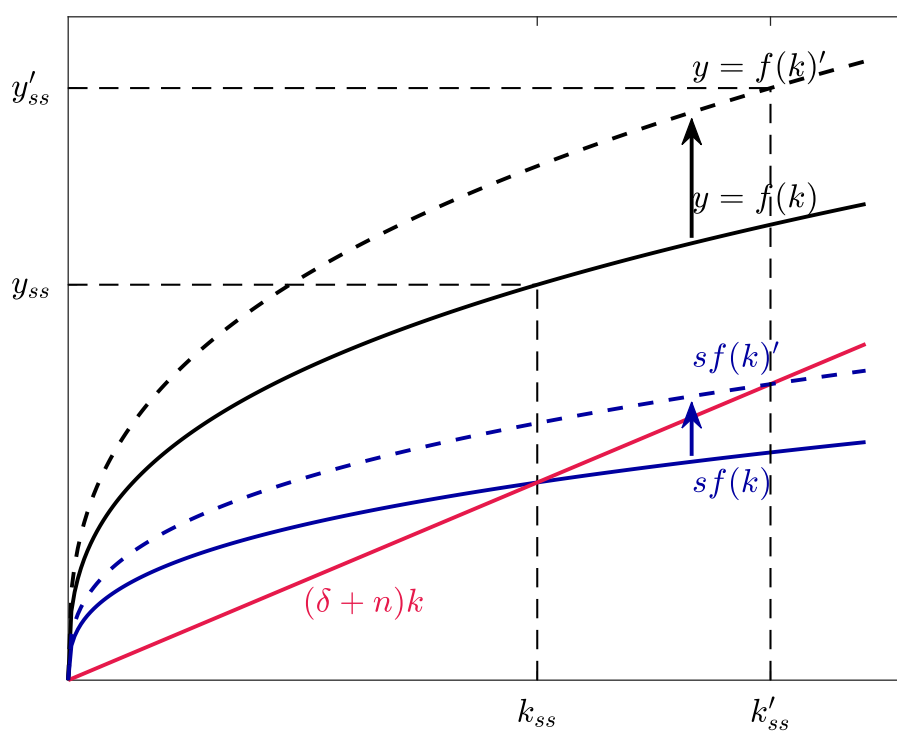
3 Growth and Level Effects: an increase in s and n



Navigation icons: back, forward, search, etc.

The Mechanics of the Solow Model

3 Growth and Level Effects: an increase in technology



Navigation icons: back, forward, search, etc.